

Itamar Avitan

PhD candidate · NeuroAI and computational cognitive neuroscience

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RESEARCH INTERESTS

How deep neural networks can reveal the computations behind biological vision and cognition, and how to tell which model is right, not only which predicts best: the identifiability of representational-alignment metrics under realistic noise (model recovery) in model-behavior and model-brain comparison; Bayesian and statistical modeling; designing experiments that separate competing models; individual differences in human visual representation.

EDUCATION

- 2024–present **PhD candidate, Industrial Engineering and Management** (direct track, integrated M.Sc.–PhD) *Ben-Gurion University of the Negev (BGU); affiliated with the School of Brain Sciences and Cognition, Israel*
- Advisor: Dr. Tal Golan, Brains and Machines Lab. Expected graduation: Sep 2029.
 - Admitted to the direct PhD track for outstanding B.Sc. graduates; PhD candidacy examination passed, 2026.
- 2026 **M.Sc. in Industrial Engineering and Management**, completed within the direct PhD track *Ben-Gurion University of the Negev, Israel*
- 2020–2024 **B.Sc. in Cognitive and Brain Sciences (computational track) and B.A. in Psychology** *Ben-Gurion University of the Negev, Israel*

PEER-REVIEWED PUBLICATION

Itamar Avitan and Tal Golan (2025). Model–Behavior Alignment under Flexible Evaluation: When the Best-Fitting Model Isn’t the Right One. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, main conference track. DOI: doi.org/10.52202/085713-0404
Preprint: arxiv.org/abs/2510.23321 | Recorded presentation: slideslive.com/39047290
Code: github.com/brainsandmachines/oddoneout_model_recovery
Data: doi.org/10.6084/m9.figshare.30542690.v1

TALKS AND PRESENTATIONS

- Sep 2026 **Flash talk** (with Maya Zach and Tal Golan), Neuro-AI-Talks (NEAT) 2026, Osnabrück, Germany. “System identification perspective on representational alignment.”
- Dec 2025 **Poster**, NeurIPS 2025, San Diego, USA, presenting the paper listed above.
- Aug 2025 **Contributed talk and poster** (with Tal Golan), Cognitive Computational Neuroscience (CCN) 2025, Amsterdam, the Netherlands. “Rethinking Representational Alignment: Linear Probing Fails to Identify the Ground-Truth Model”; preliminary version of the NeurIPS 2025 paper. Session recording: youtube.com/watch?v=vT-3kV89Rhk&t=715s

FELLOWSHIPS AND AWARDS

- 2024–present **Lachish Fellowship for Outstanding Doctoral Students**, Kreitman School of Advanced Graduate Studies, BGU (five-year award).
- 2024–present **Research Excellence Scholarship**, School of Brain Sciences and Cognition, BGU.
- 2025 **Merit-based Travel Grant**, Data Science Research Center, BGU.
- 2025 **Outstanding Poster Award**, Research Day, Department of Industrial Engineering and Management, BGU.

RESEARCH EXPERIENCE

2024–present **PhD researcher, Brains and Machines Lab**

Ben-Gurion University of the Negev, Israel

- Built a large-scale model-recovery test for representational-alignment metrics (PyTorch, Slurm GPU cluster): linearly aligned 20 diverse vision models to 4.5 million human odd-one-out judgments (THINGS dataset) and tested whether the data-generating model is re-identified from simulated responses.
- Showed that recovery accuracy plateaus below 80% even with millions of simulated trials; regression analyses linked misidentification primarily to shifts in representational geometry induced by the linear transformation. Model comparisons must therefore balance predictive accuracy against identifiability.
- Extending the approach to model–brain alignment: fitting vision networks to human functional MRI (fMRI) responses from the Natural Scenes Dataset (NSD).

PROJECTS

Jul–Aug 2026 **Embodied Brain Technology Practicum** (joint Brown University–BGU program)

Brown University, Providence, USA

- A two-week practicum for graduate students of both universities, entered by competitive selection, in which teams design, build and pitch a piece of neurotechnology. Our team of five built and pitched EarBetter, a biosignal-driven add-on for any headphones, designed to filter the sounds which trigger anxiety. Led system integration.
- Code: github.com/Itamar-Avitan/earbetter-ebt-practicum

2022–2023 **BCI4ALS**: brain–computer interface (BCI) for a person with amyotrophic lateral sclerosis (ALS)
BrainstormIL and IsrALS program, Israel

- Our team of five built an electroencephalography (EEG) P300 yes/no communication system, used live by the person it was built for. Worked on data collection, signal processing (MNE-Python preprocessing, classifier search with scikit-learn) and real-time prediction on OpenBCI hardware.
- Code: github.com/arbena/BCI4ALS

TEACHING AND MENTORING

from 2026/27 **Teaching assistant**, Academic Writing, BGU.

2023–present **Teaching assistant**, Introduction to Computation and Cognition, BGU (2023/24–2026/27).

Spring 2026 **Teaching assistant**, Deep Learning for Neuroscience and Cognition, BGU.

2025–2026 **Research mentor** to two undergraduate students in the lab (model–brain alignment; statistical modeling of neural spiking data).

Spring 2024 **Teaching assistant**, Computational Approaches to Neuroimaging, BGU.

2021–2023 **Academic tutor**, Dean of Students Office, BGU: calculus, linear algebra and statistics.

SERVICE

2025 **Reviewer**, CCN 2025, extended-abstracts track.

SKILLS AND LANGUAGES

Technical Python; PyTorch and torchvision on CUDA GPUs (deep learning); Pyro (Bayesian and probabilistic modeling); statsmodels, scikit-learn, rsatoolbox; NumPy/SciPy/pandas/xarray, Hydra; Slurm cluster computing, Git, Linux.

Languages Hebrew (native), English (fluent).